

Multiple Choice

1. **Answer:** D

Options: A) It relates inputs to outputs.; B) It is used for prediction.; C) It may be used for interpretation.; D) It discovers causal relationships

Note: Correct option: D - It discovers causal relationships

2. **Answer:** B

Options: A) The use of sampling with replacement as the sampling technique; B) The use of weak classifiers; C) The use of classification algorithms which are not prone to overfitting; D) The practice of validation performed on every classifier trained

Note: Correct option: B - The use of weak classifiers

3. **Answer:** A

Options: A) anomaly detection; B) one-class detection; C) train-test mismatch robustness; D) background detection

Note: Correct option: A - anomaly detection

4. **Answer:** B

Options: A) Increase the amount of training data.; B) Improve the optimisation algorithm being used for error minimisation.; C) Decrease the model complexity.; D) Reduce the noise in the training data.

Note: Correct option: B - Improve the optimisation algorithm being used for error minimisation.

5. **Answer:** B

Options: A) $O(1)$; B) $O(N)$; C) $O(\log N)$; D) $O(N^2)$

Note: Correct option: B - $O(N)$

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** `def find_last_occurrence(A, x): | return result`
Note: Students should provide a detailed explanation referencing algorithm steps.
12. **Answer:** `def Find_Max_Length(lst): | return maxLength`
Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key